A ROOT OF A FUNCTION IN A WEAKLY UNRAMIFIED EXTENSION OF NORMAL LOCAL DOMAINS DESCENDS TO THE SMALLER RING

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Allowed standard prerequisites (editorial).

- The local rings at height one primes of a Noetherian normal domain are discrete valuation rings, and the notion of a weakly unramified extension of such rings is the one defined for extensions of discrete valuation rings. used at proof lines 37401-37404. Makes the hypothesis of the statement meaningful and supplies the valuations used in every later step.
- Primes minimal over one element of a Noetherian domain have height one. invoked at proof lines 37405-37407. Shows that the primes minimal over the function are exactly the height one primes whose valuations the proof computes.
- An element of a normal domain whose valuation at every height one prime is bounded below by that of another element lies in the multiple of the second element. invoked at proof lines 37430-37436 and again at proof lines 37446-37448 and at the end at proof lines 37473-37476. Underlies the claim that the extension of the ideal cuts out exactly the elements with the prescribed valuation bounds, and the final comparison of the two generators.
- A module over the smaller ring which becomes free of rank one after a faithfully flat base change is itself free of rank one. invoked at proof lines 37468-37470. Descends the rank one free module property from the larger ring to the smaller one, producing the element whose power differs from the given function by a unit.

Source reference. Original definition 09E4: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32912–32912. The proof recalls at proof lines 37403-37404 that the hypothesis is the notion of that definition; the definition is a pointer fixing terminology, and every valuation computation using it is inside the counted span.

Source reference. Original lemma 00PD: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 37401-37403 to see that the local rings in question are discrete valuation rings; the valuations taken with respect to them are computed inside the counted span.

Source reference. Original lemma 00KV: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 37405-37407 to see that the primes minimal over f are

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the height one primes; the enumeration of the primes above each of them is inside the counted span.

Source reference. Original lemma 031T: sources/stacks-chow/tags.txt, source lines 1–1. Used three times inside the argument, at proof lines 37430-37436 for the injectivity claim, at proof lines 37446-37448 for the description of the extended ideal and at the end at proof lines 37473-37476 for the comparison of the two generators; each time the element whose membership is being tested is produced inside the counted span.

Source reference. Original lemma 00O1: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 37468-37470 to descend the property of being free of rank one from the larger ring to the smaller one; the identification of the extended ideal with a rank one free module is inside the counted span.

Lemma 0.2. *Let $A \to B$ be a flat local homomorphism of Noetherian local normal domains. Let $f \in A$ and $h \in B$ such that $f = wh^n$ for some $n > 1$ and some unit w of B. Assume that for every height 1 prime $\mathfrak{p} \subset A$ there is a height 1 prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p}$ such that the extension $A_\mathfrak{p} \subset B_\mathfrak{q}$ is weakly unramified. Then $f = ug^n$ for some $g \in A$ and unit u of A.*

Proof. The local rings of A and B at height 1 primes are discrete valuation rings (Algebra, Lemma 00PD). Thus the assumption makes sense (via Definition 09E4). Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the primes of A minimal over f. These have height 1 by Algebra, Lemma 00KV. For each i let $\mathfrak{q}_{i,j} \subset B$, $j = 1, \ldots, r_i$ be the height 1 primes of B lying over $\mathfrak{p}_i$. Say we number them so that $A_{\mathfrak{p}_i} \to B_{\mathfrak{q}_{i,1}}$ is weakly unramified. Since f maps to an nth power times a unit in $B_{\mathfrak{q}_{i,1}}$ we see that the valuation v_i of f in $A_{\mathfrak{p}_i}$ is divisible by n. Say $v_i = nw_i$ for some $w_i \geq 0$. Consider the exact sequence

$$0 \to I \to A \to \prod\nolimits_{i=1,\ldots,r} A_{\mathfrak{p}_i}/\mathfrak{p}_i^{w_i} A_{\mathfrak{p}_i}$$

defining the ideal I. Applying the exact functor $- \otimes_A B$ we obtain an exact sequence

$$0 \to I \otimes_A B \to B \to \prod\nolimits_{i=1,\ldots,r} (A_{\mathfrak{p}_i}/\mathfrak{p}_i^{w_i} A_{\mathfrak{p}_i}) \otimes_A B$$

Fix i. We claim that the canonical map

$$(A_{\mathfrak{p}_i}/\mathfrak{p}_i^{w_i} A_{\mathfrak{p}_i}) \otimes_A B \to \prod\nolimits_{j=1,\ldots,r_i} B_{\mathfrak{q}_{i,j}}/\mathfrak{q}_{i,j}^{e_{i,j}w_i} B_{\mathfrak{q}_{i,j}}$$

is injective. Here $e_{i,j}$ is the ramification index of $A_{\mathfrak{p}_i} \to B_{\mathfrak{q}_{i,j}}$. The claim asserts that $\mathfrak{p}_i^{w_i} B_{\mathfrak{p}_i}$ is equal to the set of elements b of $B_{\mathfrak{p}_i}$ whose valuation at $\mathfrak{q}_{i,j}$ is $\geq e_{i,j}w_i$. Choose a generator $a \in A_{\mathfrak{p}_i}$ of the principal ideal $\mathfrak{p}_i^{w_i}$. Then the valuation of a at $\mathfrak{q}_{i,j}$ is equal to $e_{i,j}w_i$. Hence, as $B_{\mathfrak{p}_i}$ is a normal domain whose height one primes are the primes $\mathfrak{q}_{i,j}$, $j = 1, \ldots, r_i$, we see that, for b as above, we have $b/a \in B_{\mathfrak{p}_i}$ by Algebra, Lemma 031T. Thus the claim.

The claim combined with the second exact sequence above determines an exact sequence

$$0 \to I \otimes_A B \to B \to \prod\nolimits_{i=1,\ldots,r} \prod\nolimits_{j=1,\ldots,r_i} B_{\mathfrak{q}_{i,j}}/\mathfrak{q}_{i,j}^{e_{i,j}w_i} B_{\mathfrak{q}_{i,j}}$$

It follows that $I \otimes_A B$ is the set of elements h' of B which have valuation $\geq e_{i,j}w_i$ at $\mathfrak{q}_{i,j}$. Since $f = wh^n$ in B we see that h has valuation $e_{i,j}w_i$ at $\mathfrak{q}_{i,j}$. Thus $h'/h \in B$ by Algebra, Lemma 031T. It follows that $I \otimes_A B$ is a free B-module of rank 1 (generated by h). Therefore I is a free A-module of rank 1, see Algebra, Lemma 00O1. Let $g \in I$ be a generator. Then we see that g and h differ by a unit in B. Working backwards we conclude that the valuation of g in $A_{\mathfrak{p}_i}$ is $w_i = v_i/n$. Hence g^n and f differ by a unit in A (by Algebra, Lemma 031T) as desired. $\square$